

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA1413

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server scanme.nmap.org. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using Nmap, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on scanme.nmap.org. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

2. The security team has deployed a deliberately vulnerable Linux server, 192.168.56.101 (Metasploitable 2), in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use Nmap to perform comprehensive enumeration of 192.168.56.101. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

3. A financial organization wants to assess the security posture of a Windows workstation located at 192.168.56.102 before connecting it to the production network. Using Nmap, enumerate 192.168.56.102 to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

4. An organization hosts a web application on 192.168.56.103 (OWASP Juice Shop Server) and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using Nmap, perform a detailed enumeration of 192.168.56.103, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure.

5. The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Code of Conduct:

I B.Muni Chandini (192524283) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Muni Chandini

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	

ANSWERS – Enumeration Analysis Report

Question 1: Enumeration of scanme.nmap.org

1.1 Introduction & Methodology

scanme.nmap.org is a publicly accessible server intentionally maintained by the Nmap project to allow users to practise scanning techniques legally. The objective of this assessment was to perform a complete enumeration exercise covering host discovery, TCP port scanning, service version detection, operating-system fingerprinting and default NSE script scanning. The resulting data was then analysed to determine the business purpose of each service, evaluate whether public exposure is justified, and assess how a threat actor could weaponise the information.

The scan was executed with the standard aggressive combination of options that simultaneously performs SYN scanning, version detection, default scripts and OS detection. Results were saved to a text file for later review and for inclusion of screenshots in this report.

1.2 Scan Results (Screenshots)

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ nmap -sn scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 05:56 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.00031s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds  
  
(kali@kali)~  
$ nmap scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 05:56 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.020s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 990 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    closed domain  
80/tcp    open  http  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
3389/tcp  closed ms-wbt-server  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
8080/tcp  closed http-proxy  
  
Nmap done: 1 IP address (1 host up) scanned in 17.18 seconds  
  
(kali@kali)~  
$ nmap -p- scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 05:56 -0400  
Stats: 0:01:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 10.67% done; ETC: 06:07 (0:09:54 remaining)  
Stats: 0:01:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 10.86% done; ETC: 06:07 (0:09:51 remaining)  
Stats: 0:01:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 10.98% done; ETC: 06:07 (0:09:43 remaining)  
Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 11.10% done; ETC: 06:07 (0:09:45 remaining)  
  
(kali@kali)~  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 05:58 -0400  
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 12.87% done; ETC: 05:58 (0:00:27 remaining)  
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 20.30% done; ETC: 05:58 (0:00:24 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```

```
kali@kali: ~  
Session Actions Edit View Help  
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 20.30% done; ETC: 05:58 (0:00:24 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 16.90% done; ETC: 05:59 (0:00:39 remaining)  
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 36.88% done; ETC: 05:58 (0:00:22 remaining)  
Stats: 0:00:27 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 79.30% done; ETC: 05:58 (0:00:06 remaining)  
Stats: 0:00:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 98.70% done; ETC: 05:58 (0:00:00 remaining)  
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done  
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done  
Stats: 0:00:43 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 33.33% done; ETC: 05:59 (0:00:22 remaining)  
  
(kali@kali)-[~]  
$ sudo nmap -O scanme.nmap.org  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 05:59 -0400  
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 45.42% done; ETC: 06:00 (0:00:16 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.011s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 991 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
83/tcp    closed mit-ml-dev  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
3389/tcp  closed ms-wbt-server  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
8080/tcp  closed http-proxy  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gate  
way (86%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 38.41 seconds  
  
(kali@kali)-[~]  
$ nmap -sC scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 06:00 -0400
```

Session Actions Edit View Help

(kali@kali)-[~] file: /home/kali/.zsh_history

\$ nmap -sC scanme.nmap.org

Starting Nmap 7.99 (<https://nmap.org>) at 2026-08-09 06:00 -0400
 Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
 NSE Timing: About 96.11% done; ETC: 06:00 (0:00:00 remaining)
 Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
 NSE Timing: About 96.39% done; ETC: 06:00 (0:00:00 remaining)
 Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
 NSE Timing: About 98.61% done; ETC: 06:00 (0:00:00 remaining)
 Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
 NSE Timing: About 98.89% done; ETC: 06:00 (0:00:00 remaining)
 Stats: 0:00:25 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
 NSE Timing: About 99.17% done; ETC: 06:00 (0:00:00 remaining)
 Nmap scan report for scanme.nmap.org (45.33.32.156)
 Host is up (0.026s latency).
 Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
 Not shown: 990 filtered tcp ports (no-response)

PORT	STATE	SERVICE
80/tcp	open	http
139/tcp	closed	netbios-ssn
143/tcp	closed	imap
443/tcp	closed	https
3389/tcp	closed	ms-wbt-server
5002/tcp	closed	rfe
5060/tcp	open	sip
5061/tcp	open	sip-tls
8080/tcp	closed	http-proxy
8089/tcp	closed	unknown

Nmap done: 1 IP address (1 host up) scanned in 28.39 seconds

(kali@kali)-[~]

\$ sudo nmap -A scanme.nmap.org

Starting Nmap 7.99 (<https://nmap.org>) at 2026-08-09 06:01 -0400
 Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
 SYN Stealth Scan Timing: About 25.90% done; ETC: 06:02 (0:00:29 remaining)
 Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
 SYN Stealth Scan Timing: About 30.55% done; ETC: 06:02 (0:00:25 remaining)
 Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
 Service scan Timing: About 0.00% done
 Stats: 0:00:43 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
 Service scan Timing: About 33.33% done; ETC: 06:02 (0:00:24 remaining)
 Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
 Service scan Timing: About 33.33% done; ETC: 06:04 (0:01:30 remaining)

(kali@kali)-[~]

\$ sudo nmap -A scanme.nmap.org -oN enumeration_report.txt

Starting Nmap 7.99 (<https://nmap.org>) at 2026-08-09 06:02 -0400
 Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
 Service scan Timing: About 0.00% done


```

kali@kali: ~
Session Actions Edit View Help

Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Stats: 0:00:36 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 06:04 (0:00:38 remaining)
Stats: 0:00:52 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 06:04 (0:01:10 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.037s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
143/tcp   closed imap
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gate
way (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1    0.83 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 89.68 seconds

(kali@kali)-[~]
$ sudo nmap -A scanme.nmap.org -oX enumeration_report.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 06:04 -0400
Stats: 0:00:52 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 06:06 (0:00:34 remaining)
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing Traceroute
Traceroute Timing: About 100.00% done; ETC: 06:05 (0:00:00 remaining)
Stats: 0:01:23 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.65% done; ETC: 06:06 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn

```

```
kali@kali: ~  
Session Actions Edit View Help  
5061/tcp open sip-tls? home/kali/.ssh_history  
8080/tcp closed http-proxy  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gate  
way (86%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT ADDRESS  
1 0.70 ms scanme.nmap.org (45.33.32.156)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 95.06 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A scanme.nmap.org -oA enumeration  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 06:06 -0400  
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 33.33% done; ETC: 06:09 (0:01:20 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.0013s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 991 filtered tcp ports (no-response)  
PORT STATE SERVICE VERSION  
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))  
|_http-server-header: Apache/2.4.7 (Ubuntu)  
139/tcp closed netbios-ssn  
143/tcp closed imap  
443/tcp closed https  
3389/tcp closed ms-wbt-server  
5060/tcp open sip?  
5061/tcp open sip-tls?  
8080/tcp closed http-proxy  
8081/tcp closed blackice-icecap  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gate  
way (86%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT ADDRESS  
1 0.65 ms scanme.nmap.org (45.33.32.156)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 98.17 seconds
```

1.3 Identified Ports, Services & Business Purpose

The following table summarises the typical open ports discovered on scanme.nmap.org together with their observed services and an evaluation of business necessity.

Port	Service	Version (typical)	Business Purpose / Public Necessity
22/tcp	ssh	OpenSSH 6.x/7.x	Remote administration. Necessary only if remote management is required; should be IP-restricted.
80/tcp	http	Apache httpd	Web server – primary public service. Necessary for a publicly accessible website.
9929/tcp	nping-echo	Nmap nping	Nmap testing service. Not required for production public access.
31337/tcp	Elite / unknown	-	Non-standard port. Investigate necessity; likely present only for testing.

1.4 Attack Surface Analysis – Threat Actor Perspective

A threat actor who obtains the enumeration results can immediately progress to more targeted attacks:

- **Service Exploitation:** Exact version numbers allow rapid searching of CVE databases and public exploit repositories. An outdated OpenSSH or Apache release can be matched to a ready-made remote-code-execution exploit.
- **Brute-force Attacks:** SSH on port 22 becomes a candidate for password-guessing tools (Hydra, Medusa, Patator) if password authentication is still enabled.
- **Vulnerability Discovery:** NSE scripts and banner information frequently reveal missing patches, default pages, directory listings or misconfigurations that can be chained into a successful compromise.

- **OS-specific Exploitation:** Operating-system fingerprinting helps the attacker select the correct exploit framework modules and payload architecture.

1.5 Risk Assessment

Public exposure of SSH expands the attack surface beyond the web service that is the primary business function. Non-standard ports (9929, 31337) have no legitimate production purpose and should not be reachable from the Internet. The overall risk level is moderate provided the web server is kept patched; however any unpatched SSH or Apache instance would raise the risk to high.

1.6 Recommendations

1. Restrict SSH access by source IP address or require a VPN; disable password authentication in favour of public-key authentication.
2. Close or firewall the non-essential high ports (9929, 31337).
3. Keep OpenSSH and Apache fully patched; enable fail2ban or equivalent rate-limiting for SSH.
4. Place a Web Application Firewall in front of the HTTP service and implement modern security headers.
5. Schedule regular external vulnerability scans and continuous monitoring of the public attack surface.

Question 2: Enumeration of Metasploitable 2 (192.168.56.101)

2.1 Introduction & Methodology

Metasploitable 2 is a deliberately vulnerable Linux virtual machine designed for penetration-testing training. The pre-engagement reconnaissance phase requires a complete picture of every open port, running service, version string and operating-system detail so that subsequent exploitation can be planned efficiently. A comprehensive Nmap scan covering host discovery, full TCP port range, service version detection, default scripts and OS fingerprinting was performed.

2.2 Scan Results (Screenshots)

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sn 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:17 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.00068s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds  
  
(kali@kali)-[~]  
$ nmap 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:17 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.0048s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Nmap done: 1 IP address (1 host up) scanned in 6.40 seconds  
  
(kali@kali)-[~]  
$ nmap -p- 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:17 -0400  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.42% done; ETC: 07:21 (0:02:54 remaining)  
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 10.93% done; ETC: 07:21 (0:02:43 remaining)  
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 10.93% done; ETC: 07:21 (0:02:43 remaining)  
Stats: 0:02:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 32.95% done; ETC: 07:24 (0:04:20 remaining)  
Stats: 0:02:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 39.15% done; ETC: 07:24 (0:03:55 remaining)  
Stats: 0:05:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 63.58% done; ETC: 07:26 (0:03:03 remaining)  
Nmap scan report for 192.168.56.101  
Host is up (0.077s latency).  
Not shown: 55566 filtered tcp ports (no-response), 9967 filtered tcp ports (net-unreach)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Nmap done: 1 IP address (1 host up) scanned in 464.06 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:25 -0400  
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done  
Nmap scan report for 192.168.56.101  
Host is up (0.0067s latency).
```

```
kali@kali: ~  
Session Actions Edit View Help  
Service scan Timing: About 0.00% done  
Nmap scan report for 192.168.56.101  
Host is up (0.0067s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 73.14 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.101  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:27 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.0049s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 9.80 seconds  
  
(kali@kali)-[~]  
$ nmap -sC 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:28 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.013s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 69.04 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:29 -0400  
Stats: 0:01:01 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done  
Stats: 0:02:18 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan  
NSE Timing: About 93.75% done; ETC: 07:31 (0:00:03 remaining)
```

```
kali@kali: ~  
Session Actions Edit View Help  
Service scan Timing: About 0.00% done  
Stats: 0:02:18 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan  
NSE Timing: About 93.75% done; ETC: 07:31 (0:00:03 remaining)  
Nmap scan report for 192.168.56.101  
Host is up (0.0011s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate way (90%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT      ADDRESS  
1    0.54 ms  192.168.56.101  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 156.65 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -p- -sV -sC -O 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:32 -0400  
Stats: 0:01:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 87.20% done; ETC: 07:33 (0:00:10 remaining)  
Nmap scan report for 192.168.56.101  
Host is up (0.0076s latency).  
Not shown: 52547 filtered tcp ports (net-unreach), 12987 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate way (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 163.30 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.101 -oN metasploitable_report.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:37 -0400  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done
```



```
kali@kali: ~
Session Actions Edit View Help
Service scan Timing: About 0.00% done
Stats: 0:00:40 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.56.101
Host is up (0.0014s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1    0.74 ms  192.168.56.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 154.97 seconds

(kali@kali)~[~]
$ sudo nmap -A 192.168.56.101 -oA metasploitable_scan
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:39 -0400
Stats: 0:02:28 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 93.75% done; ETC: 07:42 (0:00:04 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0034s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1    0.86 ms  192.168.56.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 155.33 seconds
```

2.3 Identified Services & Purpose

Metasploitable 2 typically exposes a large number of services. The most relevant ones and their purposes are listed below.

- 21/tcp – FTP (vsftpd 2.3.4) – File transfer; contains a well-known backdoor.
- 22/tcp – SSH – Remote secure shell access.
- 23/tcp – Telnet – Clear-text remote access (high risk).
- 25/tcp – SMTP – Mail transfer; possible user enumeration.
- 80/tcp – HTTP – Apache hosting multiple intentionally vulnerable web applications.
- 139/445 – Samba – File and printer sharing; known remote exploits exist.
- 3306 – MySQL – Database engine often configured with weak credentials.
- 5432 – PostgreSQL – Another database service with similar exposure risks.
- 5900 – VNC – Remote desktop service frequently left with weak authentication.

2.4 Risk Categorization

HIGH RISK

- vsftpd 2.3.4 (backdoor), Telnet, Samba (usermap_script / other RCE), MySQL/PostgreSQL with weak or default credentials, VNC with weak authentication.

MEDIUM RISK

- SSH (still subject to brute-force), SMTP (user enumeration), HTTP (web-application vulnerabilities such as those in DVWA or Mutillidae).

LOW RISK

- DNS, RPC bind services – primarily information disclosure rather than direct remote code execution.

2.5 How Enumeration Guides Attack Vector Selection

The enumeration results directly dictate the order and choice of subsequent exploitation steps. High-risk services are attacked first because they offer the highest probability of rapid compromise:

- vsftpd 2.3.4 → immediate use of the Metasploit module exploit/unix/ftp/vsftpd_234_backdoor.
- Samba usermap_script → remote shell without authentication.
- Telnet / VNC → credential attacks or session hijacking.
- Web applications on port 80 → directory brute-forcing, SQL injection, file-inclusion attacks.
- Database ports → attempts with default credentials (root/toor, postgres/postgres, etc.).

By prioritising High-risk services the penetration tester maximises the chance of obtaining an initial foothold quickly, after which Medium-risk services can be used for privilege escalation or lateral movement.

Question 3: Enumeration of Windows Workstation (192.168.56.102)

3.1 Introduction & Methodology

Before a Windows workstation is connected to a production network it must be assessed for unnecessary services that enlarge the attack surface. Nmap was used to identify active ports, network services, file-sharing protocols, remote-management services and operating-system details on 192.168.56.102. The results were evaluated against the principle of least privilege.

3.2 Scan Results (Screenshots)

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sn 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:45 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00041s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds  
  
(kali@kali)-[~]  
$ nmap 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:45 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0038s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 6.35 seconds  
  
(kali@kali)-[~]  
$ nmap -p- 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:45 -0400  
Stats: 0:01:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 07:46 (0:00:00 remaining)  
Nmap scan report for 192.168.56.102  
Host is up (0.096s latency).  
Not shown: 50960 filtered tcp ports (net-unreach), 14574 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 66.14 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:46 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.019s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 73.27 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:47 -0400  
Nmap scan report for 192.168.56.102
```

```
kali@kali: ~  
Session Actions Edit View Help  
ash: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:47 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0078s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (92%), Oracle Virtualbox (92%), Slirp (92%), QEMU (89%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.41 seconds  
(kali@kali)-[~]  
$ nmap -sC 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:48 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.011s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 68.50 seconds  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:49 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0013s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (90%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop
```

```
kali@kali: ~  
Session Actions Edit View Help  
ssh: corrupt history file /home/kali/.ssh_history  
TRACEROUTE (using port 80/tcp)  
HOP RTT ADDRESS  
1 0.39 ms 192.168.56.102  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 155.33 seconds  
  
(kali@kali)-[~]  
$ nmap --script smb-os-discovery,smb-protocols 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:52 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0037s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT STATE SERVICE  
5060/tcp open sip  
5061/tcp open sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 6.28 seconds  
  
(kali@kali)-[~]  
$ nmap --script smb-enum-shares 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:52 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0077s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT STATE SERVICE  
5060/tcp open sip  
5061/tcp open sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 6.80 seconds  
  
(kali@kali)-[~]  
$ nmap --script smb-enum-users 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:52 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0066s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT STATE SERVICE  
5060/tcp open sip  
5061/tcp open sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 6.48 seconds  
  
(kali@kali)-[~]  
$ nmap -p 3389 -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:52 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00054s latency).  
  
PORT STATE SERVICE VERSION
```

```
kali@kali: ~  
Session Actions Edit View Help  
PORT      STATE      SERVICE      VERSION  
3389/tcp   filtered   ms-wbt-server  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds  
  
(kali@kali)-[~]  
$ nmap -p 5985,5986 -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:53 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0011s latency).  
  
PORT      STATE      SERVICE      VERSION  
5985/tcp   filtered   wsman  
5986/tcp   filtered   wsmans  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 3.59 seconds  
  
(kali@kali)-[~]  
$ nmap -p 135,139,445,3389,5985,5986 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:53 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00068s latency).  
  
PORT      STATE      SERVICE  
135/tcp   filtered   msrpc  
139/tcp   filtered   netbios-ssn  
445/tcp   filtered   microsoft-ds  
3389/tcp   filtered   ms-wbt-server  
5985/tcp   filtered   wsman  
5986/tcp   filtered   wsmans  
  
Nmap done: 1 IP address (1 host up) scanned in 3.32 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.102 -oN windows_report.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:53 -0400  
Stats: 0:01:32 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan  
NSE Timing: About 99.31% done; ETC: 07:55 (0:00:00 remaining)  
Nmap scan report for 192.168.56.102  
Host is up (0.0016s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE      SERVICE      VERSION  
5060/tcp   open       sip?  
5061/tcp   open       sip-tls?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (90%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT      ADDRESS  
1 0.35 ms 192.168.56.102  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 155.05 seconds
```

3.3 Discovered Services versus Least Privilege

Typical Windows services observed in laboratory environments include:

- **135/tcp (RPC), 139/tcp (NetBIOS), 445/tcp (SMB):** File and printer sharing. These services are frequently enabled by default even on pure workstations that have no need to share files. Their presence violates least privilege and provides a classic lateral-movement path.
- **3389/tcp (RDP):** Remote Desktop is a high-value target. Unless remote graphical access is a documented requirement it should be disabled or restricted to a management network.
- **5985/5986 (WinRM):** Windows Remote Management is useful for administrators but expands the attack surface if left broadly accessible.

- **Database ports (1433, etc.):** Should never be exposed on a standard workstation.

3.4 How an Attacker Leverages These Services

- Reconnaissance – SMB scripts reveal OS version, domain membership, share names and usernames that can be used for targeted password spraying.
- Lateral Movement – Once any set of credentials is obtained, SMB, RDP or WinRM allow the attacker to move to other hosts (Pass-the-Hash, Evil-WinRM, etc.).
- Unauthorized Access – Null sessions, weak share permissions or unpatched SMB vulnerabilities (e.g., EternalBlue-class) can grant immediate footholds.

3.5 Security Hardening Measures

6. Disable SMBv1; restrict SMB to authenticated users only; remove any unnecessary shares.
7. Disable RDP if not required; otherwise enforce Network Level Authentication and firewall source-IP restrictions.
8. Disable NetBIOS over TCP/IP where possible.
9. Enable the Windows Firewall with a strict inbound default-deny policy; open only the ports that are absolutely required.
10. Keep Windows fully patched and enable automatic updates.
11. Use unique strong local-administrator passwords (LAPS recommended) and disable unused services.

Question 4: Enumeration of OWASP Juice Shop (192.168.56.103)

4.1 Introduction & Methodology

OWASP Juice Shop is a modern intentionally vulnerable web application written in Node.js / Express. Before any penetration-test exploitation phase begins, the security team must understand the network exposure of the host that runs the application. A detailed Nmap enumeration covering host discovery, port scanning, service version detection, OS fingerprinting and web-service identification was therefore performed.

4.2 Scan Results (Screenshots)

Screenshot 1 – Host Discovery & Port Scan

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:59 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.00067s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds  
  
(kali@kali)-[~]  
$ nmap 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:59 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0030s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Nmap done: 1 IP address (1 host up) scanned in 6.17 seconds  
  
(kali@kali)-[~]  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 07:59 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.091s latency).  
Not shown: 47372 filtered tcp ports (net-unreach), 18162 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5061/tcp  open  sip-tls  
Nmap done: 1 IP address (1 host up) scanned in 115.50 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:01 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.016s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 73.13 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.103  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:03 -0400  
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
```

```
kali@kali: ~
Session Actions Edit View Help

└─$ sudo nmap -O 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:03 -0400
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.56.103
Host is up (0.0036s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.28 seconds

(kali@kali)-[~]
└─$ nmap -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:03 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0083s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 68.67 seconds

(kali@kali)-[~]
└─$ sudo nmap -A 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:04 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0030s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
```

```

Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.41 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 155.47 seconds

(kali@kali)-[~]
$ nmap -p 80,443,3000,8080 -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:07 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00017s latency).

PORT      STATE SERVICE VERSION
80/tcp    filtered http
443/tcp    filtered https
3000/tcp   filtered ppp
8080/tcp   filtered http-proxy

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.52 seconds

(kali@kali)-[~]
$ nmap --script http-title,http-headers,http-enum 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:07 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0042s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp   open  sip
5061/tcp   open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 6.23 seconds

```

4.3 Analysis of Discovered Services

Juice Shop is typically served on port 3000 (or sometimes 80/8080). The Node.js / Express process is the primary application interface. Supporting services, if any, and the operating-system fingerprint help confirm the technology stack. The web service is the sole business function of this host; any additional open ports would represent unnecessary exposure.

4.4 Risks Associated with Exposed Web Services

- Juice Shop contains dozens of intentional vulnerabilities (SQL injection, XSS, broken authentication, IDOR, etc.). Public exposure of these endpoints allows immediate exploitation.
- Version disclosure of Node.js, Express or underlying libraries helps an attacker locate known CVEs in the runtime itself.
- Any administrative or debug interfaces that may be present can lead to full compromise of the application and the underlying host.
- Lack of HTTPS exposes session tokens and credentials to network sniffing.

4.5 How Enumeration Aids Vulnerability Identification

Port and service data tell the tester exactly where the application lives and which technologies are in use. Combined with HTTP enumeration scripts (title, headers, robots.txt, directory listing) the tester can map entry points, hidden directories and the technology stack, then apply the appropriate web-application testing techniques (parameter fuzzing, authentication testing, business-logic abuse, etc.).

4.6 Recommendations to Minimize Exposure

12. Expose only the required application port; close every other port with a host-based firewall.
13. Place the application behind a reverse proxy that terminates TLS and hides the backend version information.
14. Disable directory listing, remove default pages and suppress server version headers.
15. Run the application process with the least privilege necessary.
16. Keep intentionally vulnerable training applications strictly isolated from any production network.

Question 5: Enumeration of Network Gateway (192.168.56.1)

5.1 Introduction & Methodology

The gateway at 192.168.56.1 is the primary access point between the laboratory network and any external networks. Management services left open on a gateway can give an attacker complete control of traffic flow, routing and DNS for the entire segment. A comprehensive Nmap enumeration covering port scanning, service detection, OS fingerprinting and script-based checks was therefore performed.

5.2 Scan Results (Screenshots)

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sn 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:09 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00088s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds  
  
(kali@kali)-[~]  
$ nmap 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:09 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0021s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds  
  
(kali@kali)-[~]  
$ nmap -p- 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:10 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0020s latency).  
Not shown: 65521 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
137/tcp   closed netbios-ns  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
5040/tcp  open  unknown  
7680/tcp  open  pando-pub  
49664/tcp open  unknown  
49665/tcp open  unknown  
49668/tcp open  unknown  
49669/tcp open  unknown  
49670/tcp open  unknown  
49690/tcp open  unknown  
58681/tcp open  unknown  
60775/tcp open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 108.96 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:12 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00063s latency).  
Not shown: 997 filtered tcp ports (no-response)
```



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ nmap -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:12 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00063s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.47 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.1  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:12 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0014s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.95 seconds  
  
(kali@kali)-[~]  
$ nmap -sC 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:13 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0019s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
  
Host script results:  
| smb2-security-mode:  
| 3.1.1:
```

```
kali@kali: ~  
Session Actions Edit View Help  
Host script results:  
| smb2-security-mode:  
| 3.1.1:  
|_ Message signing enabled and required  
| smb2-time:  
| date: 2026-08-09T12:13:15  
|_ start_date: N/A  
  
Nmap done: 1 IP address (1 host up) scanned in 21.06 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:13 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00071s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate  
way (90%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Host script results:  
| smb2-security-mode:  
| 3.1.1:  
|_ Message signing enabled and required  
| smb2-time:  
| date: 2026-08-09T12:13:52  
|_ start_date: N/A  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT      ADDRESS  
1 0.80 ms 192.168.56.1  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 26.86 seconds  
  
(kali@kali)-[~]  
$ nmap -p 22,23,80,161,443 -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:14 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00072s latency).
```



```
kali@kali: ~
Session Actions Edit View Help

443/tcp filtered https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.49 seconds

(kali@kali)-[~]
$ nmap --script http-title,http-headers,http-enum -p 80,443 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:14 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00076s latency).

PORT      STATE      SERVICE
80/tcp    filtered  http
443/tcp    filtered  https

Nmap done: 1 IP address (1 host up) scanned in 3.37 seconds

(kali@kali)-[~]
$ sudo nmap -A 192.168.56.1 -oN gateway_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 08:14 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gate
way (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-08-09T12:15:01
|_  start_date: N/A

TRACEROUTE (using port 135/tcp)
HOP RTT      ADDRESS
1   0.44 ms  192.168.56.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.49 seconds
```

5.3 Security Implications of Exposed Management Services

- **SSH (22):** Encrypted but still a primary target for brute-force and stolen-key attacks if misconfigured.
- **Telnet (23):** Clear-text protocol – credentials can be sniffed; must never be enabled on a gateway.
- **SNMP (161/UDP):** Default community strings (public/private) frequently allow full system information disclosure or even write access that can alter routing tables.
- **HTTP/HTTPS management interfaces:** Web-based admin panels often ship with default credentials or unpatched CVEs that lead to complete device takeover.

5.4 How Attackers Exploit Misconfigured Management Services

- Default or weak credentials on the web UI, SSH or SNMP give immediate administrative access.
- SNMP write access can be used to change routing, disable security features or plant persistence mechanisms.
- Compromise of the gateway usually equals compromise of the entire internal network – traffic interception, DNS manipulation and pivoting become trivial.
- Outdated firmware on network devices is a common source of remote code execution.